# Network Investigation

## Exercise: 1.4 | External Network Communication Detected

### Objectives:

The primary objective is to conduct a thorough investigation into a suspected external network communication Incident with the goal of identifying the source, scope, and impact of the attack.

### Incident Details:

Investigate the suspicious external communication activity detected by the network monitoring systems. Investigate the PCAP to identify the targeted IP, source, and potential impact.

### PCAP File Name:

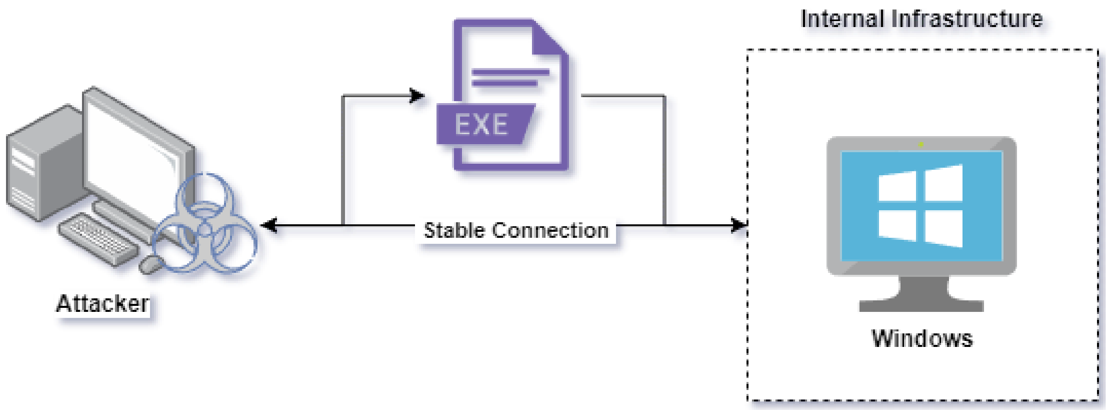
| |
|---|
| external.pcapng |

### Targeted Port:

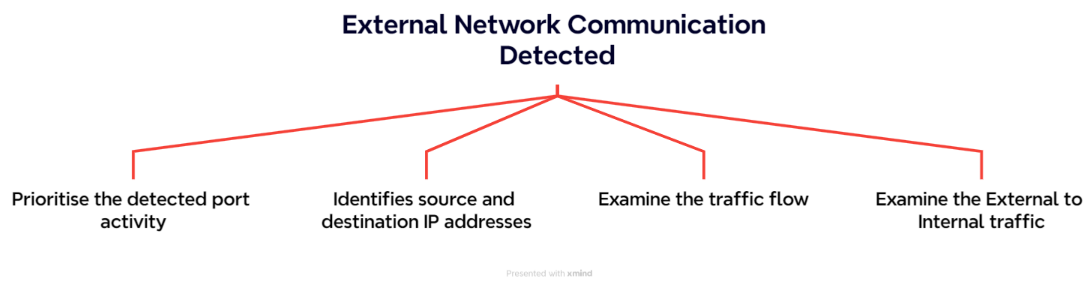
| |
|---|
| 444 |

### About RFI:

External network communication refers to the interaction and data exchange between a local network and external networks, when a attacker successfully deploy the payload over the host machine it will establish a stable connection between the attacker and the victim



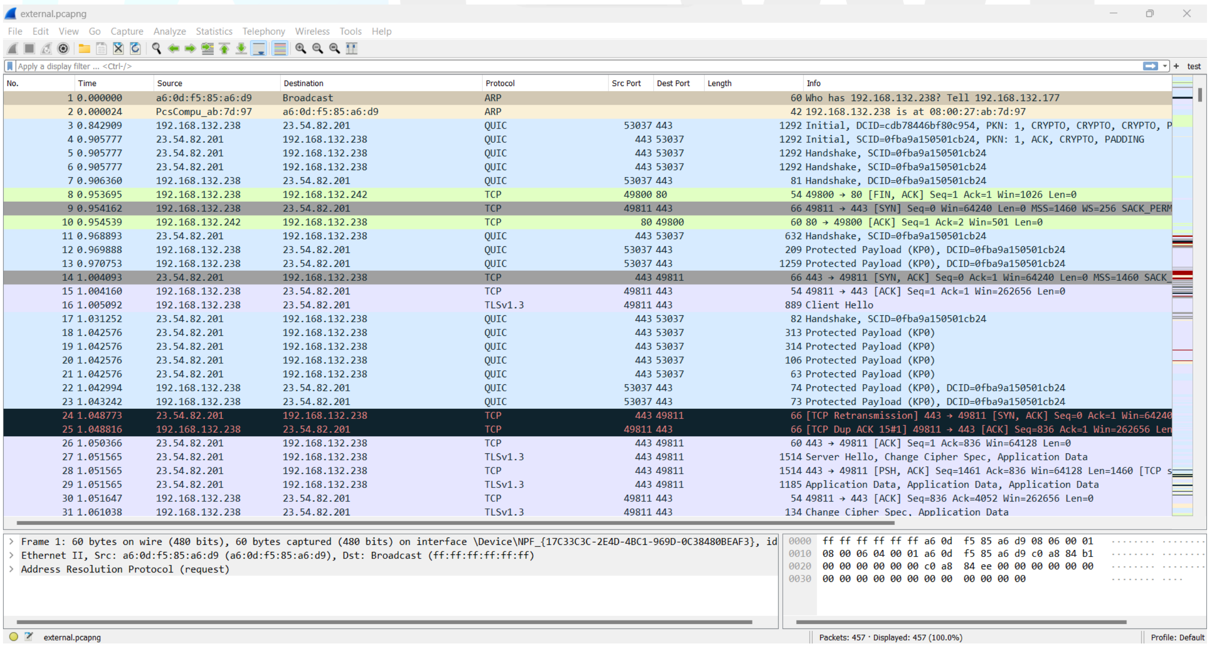
## Investigation Mindmap:




### Step:01

The first step of the investigation begins by analysing the given network PCAP. We have provided you with the suspected PCAP file which has been extracted from one of the compromised host machines.

Download the given PCAP dump on your own host machine for further investigation.




**Wireshark** is a popular open-source network protocol analyzer that allows users to capture and inspect the data travelling back and forth on a network in real time

**Step:03**

Investigating external communication involves careful steps to ensure security and determine the nature and potential threats associated with the network logs.

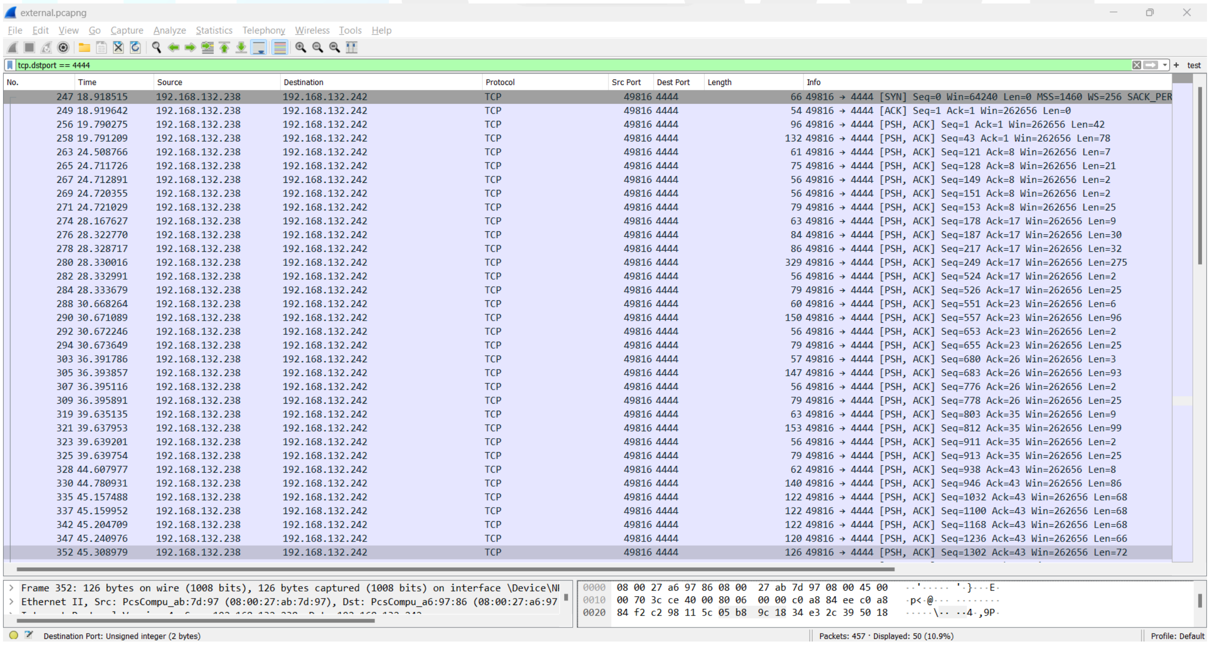
1. Prioritise the detected port activity
2. Identifies source and destination IP addresses
3. Examine the traffic flow
4. Examine the External to Internal traffic

## Prioritise the detected port activity

Identifying and extracting particular logs which are associated with the identified activity

Use the following command to obtain the web-access log.

```
Unset
tcp.dstport == 4444
```



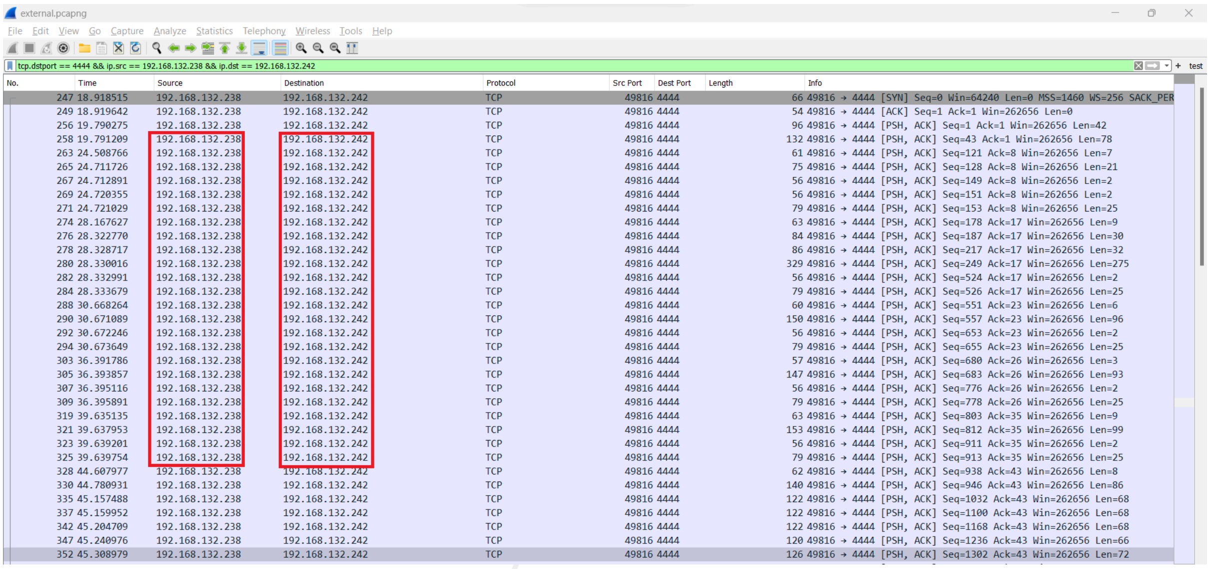

> **Decoders** are the key components which are responsible for parsing raw logs into a structured format that can be easily analysed and understood by the system.

### Identifies source and destination IP addresses

The next step is to identify and retrieve the IP activity which are associated with the detected port activity, from the above result we identified that the src 192.168.132.238 → 192.168.132.242 over destination port 4444

Unset

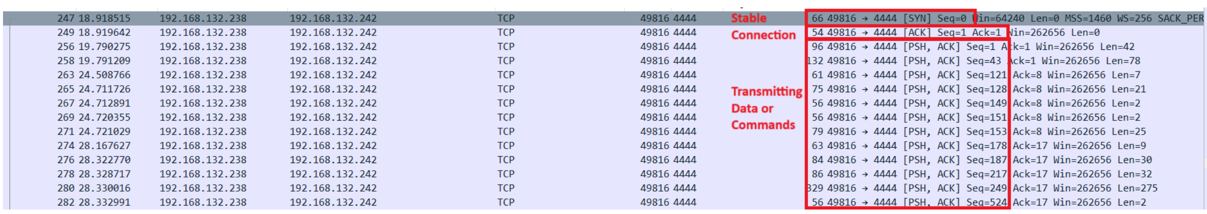
```
tcp.dstport == 4444 && ip.src == 192.168.132.238 && ip.dst ==
192.168.132.242
```




| No. | Time | Source | Destination | Protocol | Src Port | Dest Port | Length | Info |
|---|---|---|---|---|---|---|---|---|
| 247 | 18.918515 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 66 | 49816 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PER |
| 249 | 18.919642 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 54 | 49816 → 4444 [ACK] Seq=1 Ack=1 Win=262656 Len=0 |
| 256 | 19.790275 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 96 | 49816 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=42 |
| 258 | 19.791209 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 132 | 49816 → 4444 [PSH, ACK] Seq=43 Ack=1 Win=262656 Len=78 |
| 263 | 24.508766 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 61 | 49816 → 4444 [PSH, ACK] Seq=121 Ack=8 Win=262656 Len=7 |
| 265 | 24.711726 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 75 | 49816 → 4444 [PSH, ACK] Seq=128 Ack=8 Win=262656 Len=21 |
| 267 | 24.712891 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=149 Ack=8 Win=262656 Len=2 |
| 269 | 24.720355 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=151 Ack=8 Win=262656 Len=2 |
| 271 | 24.721029 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=153 Ack=8 Win=262656 Len=25 |
| 274 | 28.167627 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 63 | 49816 → 4444 [PSH, ACK] Seq=178 Ack=17 Win=262656 Len=9 |
| 276 | 28.322770 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 84 | 49816 → 4444 [PSH, ACK] Seq=187 Ack=17 Win=262656 Len=30 |
| 278 | 28.328717 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 86 | 49816 → 4444 [PSH, ACK] Seq=217 Ack=17 Win=262656 Len=32 |
| 280 | 28.330016 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 329 | 49816 → 4444 [PSH, ACK] Seq=249 Ack=17 Win=262656 Len=275 |
| 282 | 28.332991 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=524 Ack=17 Win=262656 Len=2 |
| 284 | 28.333679 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=526 Ack=17 Win=262656 Len=25 |
| 288 | 30.668264 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 60 | 49816 → 4444 [PSH, ACK] Seq=551 Ack=23 Win=262656 Len=6 |
| 290 | 30.671089 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 150 | 49816 → 4444 [PSH, ACK] Seq=557 Ack=23 Win=262656 Len=96 |
| 292 | 30.672246 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=653 Ack=23 Win=262656 Len=2 |
| 294 | 30.673649 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=655 Ack=23 Win=262656 Len=25 |
| 303 | 36.391786 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 57 | 49816 → 4444 [PSH, ACK] Seq=680 Ack=26 Win=262656 Len=3 |
| 305 | 36.393857 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 147 | 49816 → 4444 [PSH, ACK] Seq=683 Ack=26 Win=262656 Len=93 |
| 307 | 36.395116 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=776 Ack=26 Win=262656 Len=2 |
| 309 | 36.395891 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=778 Ack=26 Win=262656 Len=25 |
| 319 | 39.635135 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 63 | 49816 → 4444 [PSH, ACK] Seq=803 Ack=35 Win=262656 Len=9 |
| 321 | 39.637953 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 153 | 49816 → 4444 [PSH, ACK] Seq=812 Ack=35 Win=262656 Len=99 |
| 323 | 39.639201 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=911 Ack=35 Win=262656 Len=2 |
| 325 | 39.639754 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=913 Ack=35 Win=262656 Len=25 |
| 328 | 44.607977 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 62 | 49816 → 4444 [PSH, ACK] Seq=938 Ack=43 Win=262656 Len=8 |
| 330 | 44.780931 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 140 | 49816 → 4444 [PSH, ACK] Seq=946 Ack=43 Win=262656 Len=86 |
| 335 | 45.157488 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 122 | 49816 → 4444 [PSH, ACK] Seq=1032 Ack=43 Win=262656 Len=68 |
| 337 | 45.159952 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 122 | 49816 → 4444 [PSH, ACK] Seq=1100 Ack=43 Win=262656 Len=68 |
| 342 | 45.204709 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 122 | 49816 → 4444 [PSH, ACK] Seq=1168 Ack=43 Win=262656 Len=68 |
| 347 | 45.240976 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 120 | 49816 → 4444 [PSH, ACK] Seq=1236 Ack=43 Win=262656 Len=66 |
| 352 | 45.308979 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 126 | 49816 → 4444 [PSH, ACK] Seq=1302 Ack=43 Win=262656 Len=72 |

### Examine the traffic flow

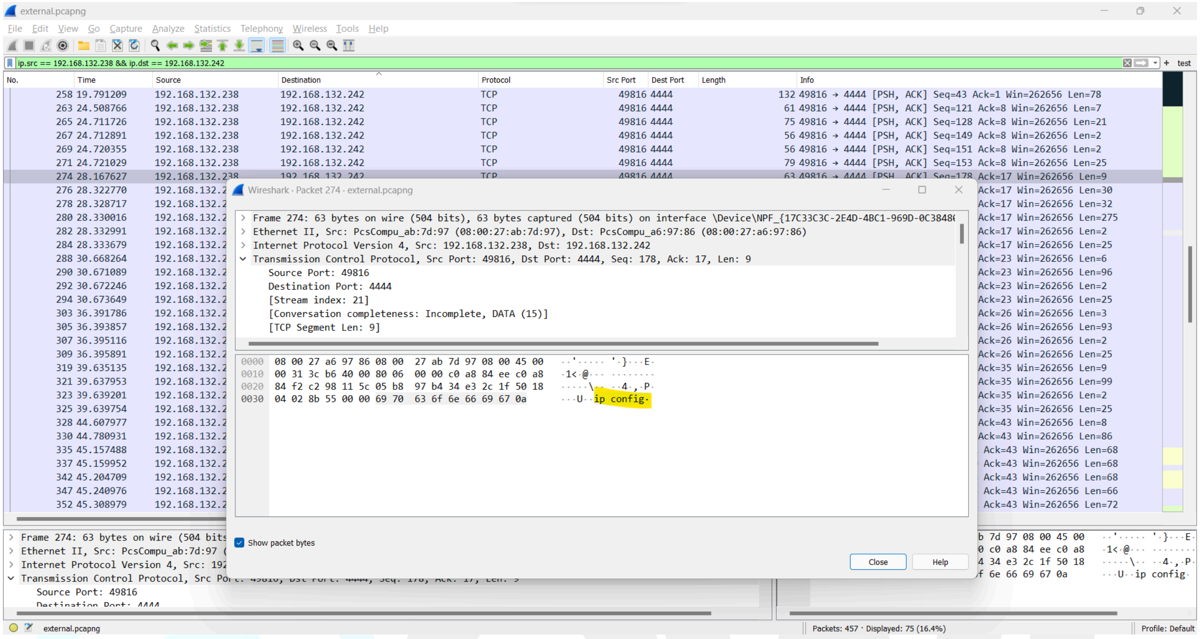
To examine the network traffic it is recommended to check the packet flag from the above observation result. We identified **SYN ACK** this pattern observed when a src dest established a stable communication, and **PSH** command generally observed when a src trying to transmit data or executing command.

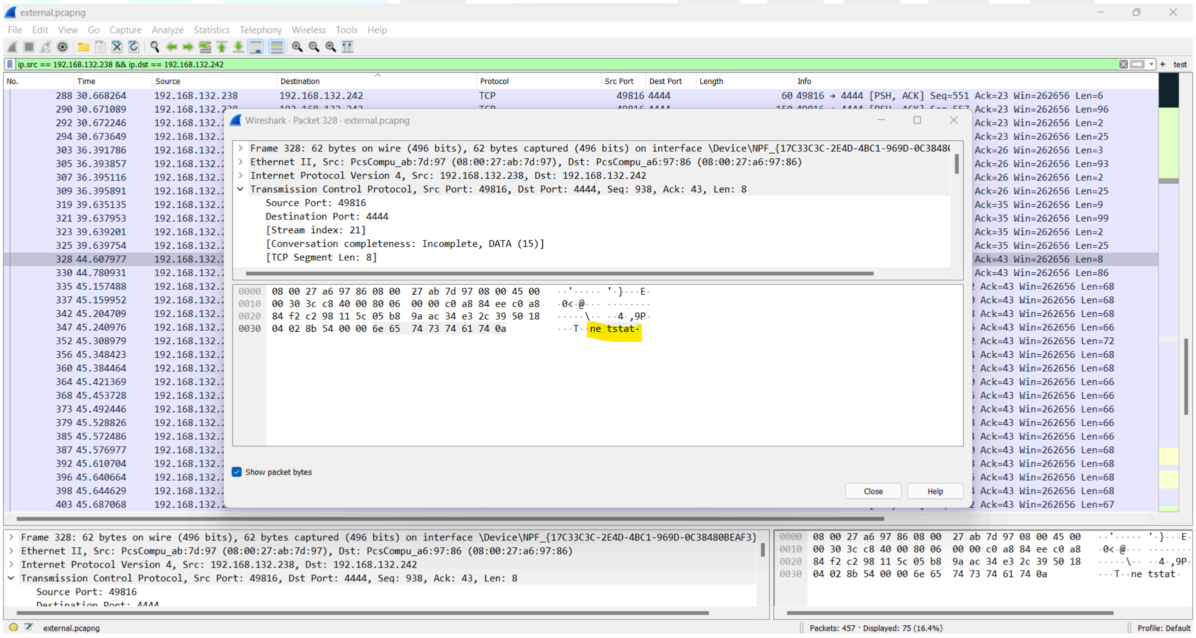



| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 247 | 18.918515 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 66 | 49816 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PER |
| 249 | 18.919642 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 54 | 49816 → 4444 [ACK] Seq=1 Ack=1 Win=262656 Len=0 |
| 256 | 19.790275 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 96 | 49816 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=42 |
| 258 | 19.791209 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 132 | 49816 → 4444 [PSH, ACK] Seq=43 Ack=1 Win=262656 Len=78 |
| 263 | 24.508766 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 61 | 49816 → 4444 [PSH, ACK] Seq=121 Ack=8 Win=262656 Len=7 |
| 265 | 24.711726 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 75 | 49816 → 4444 [PSH, ACK] Seq=128 Ack=8 Win=262656 Len=21 |
| 267 | 24.712891 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=149 Ack=8 Win=262656 Len=2 |
| 269 | 24.720355 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=151 Ack=8 Win=262656 Len=2 |
| 271 | 24.721029 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 79 | 49816 → 4444 [PSH, ACK] Seq=153 Ack=8 Win=262656 Len=25 |
| 274 | 28.167627 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 63 | 49816 → 4444 [PSH, ACK] Seq=178 Ack=17 Win=262656 Len=9 |
| 276 | 28.322770 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 84 | 49816 → 4444 [PSH, ACK] Seq=187 Ack=17 Win=262656 Len=30 |
| 278 | 28.328717 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 86 | 49816 → 4444 [PSH, ACK] Seq=217 Ack=17 Win=262656 Len=32 |
| 280 | 28.330016 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 329 | 49816 → 4444 [PSH, ACK] Seq=249 Ack=17 Win=262656 Len=275 |
| 282 | 28.332991 | 192.168.132.238 | 192.168.132.242 | TCP | 49816 | 4444 | 56 | 49816 → 4444 [PSH, ACK] Seq=524 Ack=17 Win=262656 Len=2 |

This particular pattern was observed when a victim got compromised and established a stable shell with the attacker machine.

On further checking the packet we identified multiple os command associated with the network activity

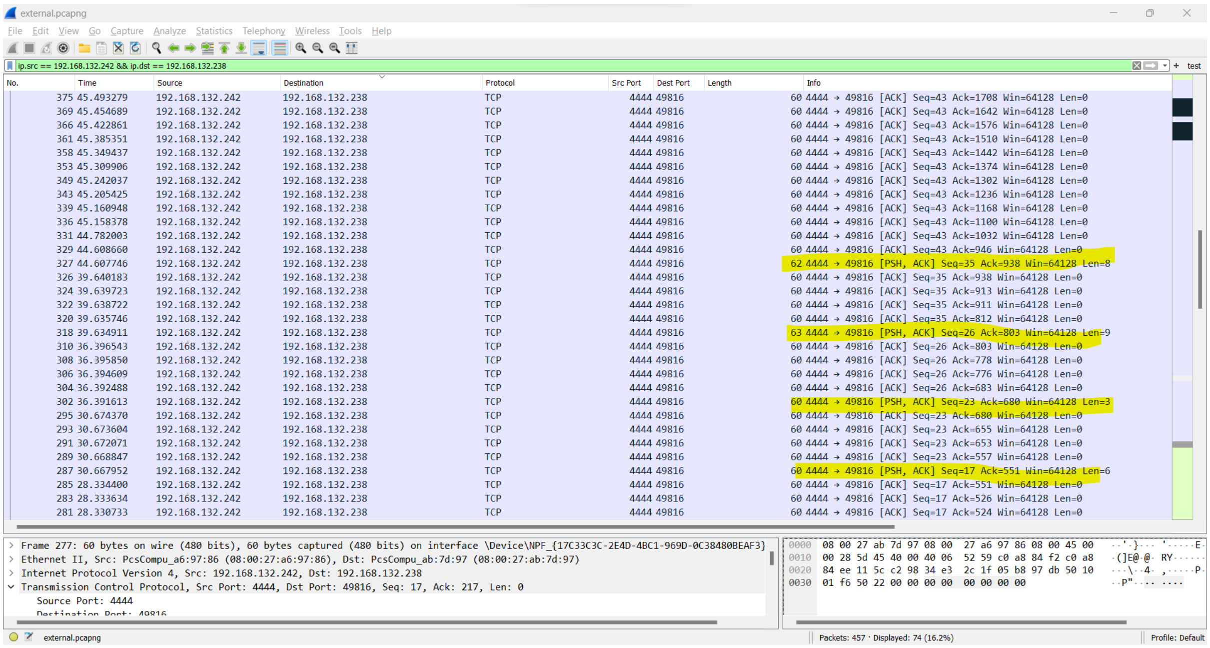


### Examine the External to Internal traffic

After determining the suspicious attack pattern its recommend to check the reverse traffic vice verse, by examin the vice verse traffic its give a better visibility over the detected activity

```Unset
ip.src == 192.168.132.242 && ip.dst == 192.168.132.238
```




As per the above result we identified a stable communication with multiple PUSH requests, examining the PUSH request to identify the command execution.

### Conclusion

The thorough examination of the incident has allowed you to understand the sequence of events leading up to the occurrence, identifying areas for improvement in processes and procedures.

# Thank You!

Hope you have learnt a lot after practically completing
Blue Team Fundamentals Exercise.
If you like the investigation & the lab scenario,
feel free to tag us & share your achievement / progress
on social media platforms :)

For Enterprise Red Team / Blue Team / Purple Team
Cyber Range Labs / Training please email :
support@cyberwarfare.live

To know more visit at : https://cyberwarfare.live